

VOLUME BARS AND LINES

Pine Script v6 | Liquidity-Tier Volume Indicator

User Manual & Mathematical Reference



10-tier volume colour scale | grey → light-blue → blue → green → yellow → red → white

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Built on principles of relative volume analysis, true exponential mathematics,
liquidity engineering, and statistical significance testing.

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1. Indicator Overview & Philosophy

Volume Bars and Lines is a liquidity-tier colouring and structural pivot indicator for TradingView Pine Script v6. Its core premise is that **volume is the footprint of institutional money**: every significant price level that acts as future support or resistance was created at a moment of abnormally high transactional activity. By quantifying and colour-coding that activity, the trader can see — at a glance — which bars carry genuine market conviction and which are low-participation noise.

The indicator operates on three interlocking layers:

- **Bar colouring** — each candlestick is recoloured according to its volume tier (0–9), giving an immediate visual heat map of market participation across time.
 - **Heikin-Ashi wick overlay** — smoothed HA wicks are rendered in the same tier colour, reducing price noise while preserving directional information.
 - **Liquidity pivot lines** — when a pivot high or low forms on a bar whose volume meets the minimum tier threshold, a horizontal line is extended to the right. These lines mark genuine liquidity nodes where unfilled orders are likely resting.
- *The indicator does NOT repaint. Pivot lines are drawn only after the full confirmation window ($2 \times \text{Pivot Length bars}$) has closed.*

2. Mathematical Foundations

2.1 Relative Volume (RVOL) Normalisation

Raw volume figures are instrument-specific and timeframe-specific: a volume of 1,000 is enormous for a thinly-traded small-cap but trivial for BTCUSDT. Manual calibration via Magnitude and Multiplier is therefore fragile and requires adjustment for every instrument. Relative Volume solves this by expressing each bar's volume as a multiple of its own rolling average:

$$RVOL(t) = V(t) / SMA(V, N)(t)$$

where $V(t)$ is the raw volume at bar t and N is the lookback period (default 50). An RVOL of 1.0 means exactly average volume. An RVOL of 3.0 means three times the recent average. This ratio is then compared against the scale thresholds $t_0 \dots t_9$ to assign a colour tier:

$$tier(t) = \max \{ i \in \{0, \dots, 9\} : RVOL(t) \geq t_i \}$$

The SMA lookback N controls sensitivity. A shorter window (20–30) adapts quickly to regime changes in volume, while a longer window (100–200) provides a more stable baseline, making outliers more visually prominent.

2.2 Scale Types

The nine threshold multipliers $t_0 \dots t_9$ define the boundaries between colour tiers. The choice of scale determines how sensitively the indicator responds to volume changes. Five mathematically distinct progressions are available:

Linear — $t_i = i$

Uniform steps of $1 \times RVOL$ per tier. Appropriate for instruments with stable, low-variance volume distributions. Provides maximum tier resolution at low multiples but compresses the upper tiers. Mathematically: an arithmetic sequence.

Quadratic (original) — $t_i = i^2$

The original script's 'exponential' mode was in fact a quadratic (polynomial of degree 2) sequence: 1, 4, 9, 16, 25, 36, 49, 64, 81. This provides moderate separation between lower tiers and aggressive compression of higher tiers. Growth rate $d^2 t_i / di^2 = 2$ (constant second derivative), characteristic of a parabola.

True Exponential — $t_i = b^{i-1}$

Each successive threshold is b times the previous: 1, b , b^2 , b^3 , ..., b^i . With $b = 2$ the thresholds are 1, 2, 4, 8, 16, 32, 64, 128, $256 \times RVOL$. This is the mathematically correct exponential scale — the ratio between consecutive thresholds is constant (equal to b), which matches the log-normal distribution of financial volume. This is the **recommended default**.

$$t_i = b^{(i-1)} \text{ ratio: } t_{i+1}/t_i = b \text{ (constant)}$$

Fibonacci — $t_i \in \{1, 2, 3, 5, 8, 13, 21, 34, 55\}$

Uses the Fibonacci sequence as thresholds. The golden ratio $\phi \approx 1.618$ governs the growth rate of consecutive Fibonacci numbers asymptotically. This scale has denser resolution in the lower tiers ($1-8\times$) and sparser resolution above, reflecting the natural clustering of volume events in financial markets.

Log-Spaced — $t_i = 10^{(i/8)}$

Eight equally-spaced points on a base-10 logarithmic axis from 1 to 10: 1.00, 1.33, 1.78, 2.37, 3.16, 4.22, 5.62, 7.50, 10.00. This scale is perceptually uniform — equal visual distance on a log chart corresponds to equal tier width. Useful when the analyst wants symmetric tier separation on a logarithmic price chart.

Scale Threshold Comparison Table

Tier	Linear	Quadratic	True Exp (b=2)	Fibonacci	Log-Spaced
0	1	1	1	1	1.0
1	2	4	2	2	1.33

Tier	Linear	Quadratic	True Exp (b=2)	Fibonacci	Log-Spaced
2	3	9	4	3	1.78
3	4	16	8	5	2.37
4	5	25	16	8	3.16
5	6	36	32	13	4.22
6	7	49	64	21	5.62
7	8	64	128	34	7.5
8	9	81	256	55	10.0

All values are RVOL multiples (i.e., 4× = four times average volume). True Exponential with b=2 is the theoretically optimal choice for log-normally distributed volume data.

2.3 Volume Z-Score & Statistical Significance

Beyond relative ranking, a statistically grounded measure of volume exceptionality is the Z-score — the number of standard deviations a bar's volume lies above the rolling mean:

$$Z(t) = (V(t) - \mu(t)) / \sigma(t)$$

where $\mu(t) = \text{SMA}(V, N)(t)$ and $\sigma(t) = \text{StdDev}(V, N)(t)$ use the same lookback N as RVOL . Under the assumption that log-volume is approximately normally distributed:

- $|Z| > 2 \rightarrow$ volume exceeds mean by 2σ ; occurs in $\sim 2.3\%$ of bars (yellow background tint)
- $|Z| > 3 \rightarrow$ volume exceeds mean by 3σ ; occurs in $\sim 0.13\%$ of bars (white background tint)
- These events correspond to earnings releases, macro announcements, liquidation cascades, and other genuine market-structure-altering events.
- **Z-score assumes approximate normality of volume. In crypto markets, volume distributions are heavily right-skewed (fat tails), so $Z > 3$ events occur more frequently than the Gaussian model predicts. Use Z-score tinting as a supplementary signal, not a primary filter.**

2.4 Tier Assignment Algorithm

The complete tier assignment logic in pseudocode:

```
avgVol ← SMA(volume, N) baseUnit ← avgVol // RVOL mode OR 10^(mag-1) × mult // legacy mode volNorm ←
volume / max(baseUnit, ε) tier ← 0 if volNorm < t1 1 if t1 ≤ volNorm < t2 2 if t2 ≤ volNorm < t3 ... 9
if volNorm ≥ t9 colour ← colourMap[tier]
```

The thresholds $t_1 \dots t_9$ are computed once per bar from the selected scale type and base, then compared against the normalised volume. The assignment is a simple ordered lookup with $O(1)$ complexity per bar — there is no loop over tiers at runtime.

3. Volume Colour Tier Reference

The ten colour tiers form a continuous spectrum from market inactivity (grey) through moderate participation (blue / green) to extreme liquidity events (red / white). The thresholds below assume True Exponential scale with base 2 and RVOL mode:

Tier	Swatch	Name	RVOL Threshold (base 2)	Market Interpretation
0		Below threshold	$< 1\times$	Inactive / no participation
1		Light blue — Low	$\geq 1\times$	Mild above-average interest
2		Blue — Moderate	$\geq 2\times$	Notable liquidity building
3		Light green — Elevated	$\geq 4\times$	Institutional accumulation signal
4		Green — High	$\geq 8\times$	Strong directional commitment
5		Light yellow — Very high	$\geq 16\times$	Momentum inflection zone
6		Yellow — Extreme	$\geq 32\times$	Climax / exhaustion watch
7		Light red — Alert	$\geq 64\times$	Liquidity vacuum risk
8		Red — Critical	$\geq 128\times$	Macro event / forced liquidation
9		White — Climactic	$\geq 256\times$	Historic event volume

Thresholds shown are RVOL multiples for True Exponential scale $b=2$. Under other scales the RVOL values differ — see Section 2.2.

The colour progression follows the visible light spectrum in reverse — cold colours (blue) for moderate activity, warm colours (yellow → red) for high activity, and white for the most extreme events. This is consistent with heat-map conventions used in institutional order-flow analysis tools.

4. Liquidity Pivot Detection

4.1 Pivot Mathematics

A pivot high at bar index p (i.e., $\text{bar_index}[L]$ where $L = \text{Pivot Length}$) is defined as:

$$\text{PivotHigh}(p, L) \equiv \forall i \in [p-L, p-1]: \text{high}[i] < \text{high}[p] \wedge \forall i \in [p+1, p+L]: \text{high}[i] \leq \text{high}[p]$$

The left window uses a strict inequality (no bar may exceed the pivot), while the right window uses a non-strict inequality (bars may equal but not exceed). This asymmetry prevents double-counting of plateau tops and ensures the most recent occurrence of a repeated high is selected.

Symmetrically, a pivot low at bar p is:

$$\text{PivotLow}(p, L) \equiv \forall i \in [p-L, p-1]: \text{low}[i] > \text{low}[p] \wedge \forall i \in [p+1, p+L]: \text{low}[i] \geq \text{low}[p]$$

The total detection window is $2L$ bars. Increasing L produces fewer, more structurally significant pivots. Decreasing L produces more frequent, shorter-term pivots. A value of $L = 4-8$ is appropriate for swing trading; $L = 1-2$ for scalping.

■ **Pivots are confirmed only after $2L$ bars have elapsed since the candidate bar. Lines therefore appear L bars 'late' relative to the pivot price level. This is non-repainting by design.**

4.2 Volume-Tier Filtering

Not all structural pivots are meaningful liquidity nodes. A pivot formed on below-average volume indicates that the price level was created without genuine institutional participation — it is unlikely to act as a strong support or resistance in the future. The Min Volume Tier parameter filters out these low-quality pivots:

$$\text{DrawLine}(p) \equiv \text{PivotConfirmed}(p) \wedge \text{Tier}(p) \geq \text{MinTier}$$

Recommended values for MinTier:

- **0** — Draw all pivots regardless of volume (noisy; useful for debugging or scalping)
- **2** — Blue tier and above: moderate liquidity filter (default)
- **4** — Green tier and above: institutional-grade pivots only
- **5+** — Yellow and above: extreme events only (very sparse; macro timeframes)

4.3 RVOL Label Annotation

Each pivot line is annotated with a label displaying the raw volume and, when RVOL mode is active, the RVOL ratio at the pivot bar:

Label: "1 234 567 (3.2×)"

The raw volume provides an absolute reference for cross-timeframe comparison. The RVOL multiple immediately communicates how exceptional the event was relative to the instrument's own history at that time — a 3.2× event means the pivot formed on volume 3.2 times the 50-bar average, regardless of whether that average is 100 or 10,000,000 contracts.

5. Input Parameter Reference

Parameter	Type	Default	Description
USE RELATIVE VOLUME (RVOL)	bool	true	Normalise volume against rolling SMA. Eliminates per-instrument manual calibration. Recommended ON for all assets.
RVOL Lookback Period	int	50	Window length N for SMA(volume,N). Shorter = more reactive. Longer = more stable baseline. 20–100 covers most timeframes.
Tint Background by Z-Score	bool	false	Overlays yellow (Z>2) or white (Z>3) background shading to flag statistically extreme volume events.
Magnitude	int 1–9	3	Legacy only (RVOL OFF). Sets order of magnitude: 1=×1, 2=×10, ... 9=×100M. Match to typical volume of the instrument.
Multiplier	int 1–9	2	Legacy only. Fine-tunes base unit within the chosen magnitude. base = 10 ^(mag–1) × mult.
Scale Type	string	True Exp	Mathematical spacing of the 9 volume tiers. See Section 3 for full comparison. True Exponential recommended.
Exponential Base	float	2.0	Base b for True Exponential scale. $t_i = b^{(i-1)}$. b=2 → each tier requires double the volume of previous.
Time Frame	resolution	(chart)	Override timeframe for Heikin-Ashi OHLC data used in wick plotting and pivot detection.
Pivot Length	int	4	Symmetrical window for pivot detection. A pivot high at bar[L] requires no bar in [0,L–1] higher, and none in [L+1,2L] equal-or-higher.
Show Lines	bool	true	Toggle all pivot liquidity lines on/off.
Show Labels	bool	true	Toggle volume annotation labels on pivot lines.
Min Volume Tier (0–9)	int	2	Only draw pivot lines where the pivot bar's volume tier ≥ this value. Eliminates low-liquidity noise pivots. Set 4–5 for cleaner charts.
High/Low Line Style	string	dotted	Visual style of drawn lines: solid, dotted, dashed, or arrow variants.
Show Heikin-Ashi Wick	bool	true	Render HA-smoothed wicks colour-matched to current volume tier, revealing candle direction without noise.

6. Recommended Configurations

6.1 Cryptocurrency (BTC / ETH / Major Alts)

Parameter	Recommended Value
Use RVOL	ON
RVOL Lookback	50 (daily) / 20 (4H)
Scale Type	True Exponential
Exp Base	2.0
Pivot Length	5–8 (daily) / 3–4 (4H)
Min Volume Tier	3 (green)
Z-Score Tint	ON — crypto has frequent outliers

Crypto volume distributions are highly right-skewed due to leverage liquidations and exchange-specific flow. RVOL with True Exponential $b=2$ handles this well. Enable Z-Score tinting to identify liquidation cascades ($Z > 3$ = white tint).

6.2 Equities & Futures (ES, NQ, SPY, Single Stocks)

Parameter	Recommended Value
Use RVOL	ON
RVOL Lookback	50–100 (aligns with ~2 trading months)
Scale Type	True Exponential OR Fibonacci
Exp Base	1.5–2.0
Pivot Length	4–6
Min Volume Tier	2–3
Z-Score Tint	ON for earnings plays; OFF otherwise

Equity volume follows cleaner log-normal distributions than crypto. Fibonacci scale works well because institutional order size tends to cluster at Fibonacci multiples of ADTV. The RVOL lookback should span at least one full earnings cycle (50–100 trading days).

6.3 Forex & Low-Volume / Illiquid Assets

Parameter	Recommended Value
Use RVOL	ON (essential — no absolute volume baseline possible)
RVOL Lookback	100–200 (very stable baseline needed)
Scale Type	Log-Spaced OR Linear
Pivot Length	3–5

Parameter	Recommended Value
Min Volume Tier	1–2 (volume is inherently lower)
Z-Score Tint	ON — highlights the rare high-volume FX events

Forex tick volume is a proxy for true volume. Absolute calibration is meaningless. RVOL is mandatory. Use a long lookback (100–200) to establish a stable baseline across multiple sessions and DST transitions. Log-Spaced scale provides finer resolution in the lower RVOL ranges typical of FX.

7. Liquidity Engineering Interpretation Guide

Volume is the mechanism by which price levels acquire significance. A price level that traded on high volume has a large inventory of filled orders at that level — both from participants who profited and those who are offside. Both groups create future order flow at and around that price:

- **Profitable longs at a high-volume low** will add to positions on retests (demand zone reinforcement)
- **Breakeven / losing longs at a high-volume high that reversed** will sell on retests to exit flat (supply zone / resistance)
- **Short sellers who sold into a high-volume spike** will cover on any revisit (buy-side demand on pullback)

Tier-Based Market Structure Interpretation

- ■ **Tier 0 (grey):** Market is in low-participation drift. No directional commitment. Avoid entries — any move is easily reversed.
- ■ **Tier 1–2 (light/mid blue):** Baseline participation. Normal trending conditions. Standard S/R analysis applies.
- ■ **Tier 3–4 (green):** Institutional accumulation or distribution. Pivots at these tiers are high-probability S/R levels. Monitor for follow-through.
- ■ **Tier 5–6 (yellow):** Momentum-driven participation. Often accompanies breakouts from consolidation or news-driven moves. Lines drawn here are strong magnet levels for future price action.
- ■ **Tier 7–8 (red):** Potential exhaustion / climax. High probability of mean reversion following a tier-8 bar. If price reverses at a tier-8 bar, the resulting pivot line is an extremely high-conviction level.
- ■ **Tier 9 (white):** Historic event. Black swan / liquidation cascade / macro catalyst. These levels are permanent structural references. Price will often return to the Tier-9 bar's high or low years later.

Confluence Framework

The highest-probability trades occur when multiple signals converge at the same price level:

Signal	Condition	Conviction
Tier-4+ pivot line	Price approaches a green-tier pivot level	HIGH
Multi-tier pivot cluster	Two or more pivot lines within 0.1% of each other	VERY HIGH
Z-Score tint + pivot	Z>2 bar coincides with confirmed pivot	VERY HIGH
Tier escalation	3+ consecutive rising-tier bars in a trend	MODERATE-HIGH
Tier collapse	Tier drops from 5+ to 0–1 after a spike	EXHAUSTION SIGNAL

8. Mathematical Appendix

A. Scale Type Formulae

Scale	Formula	Growth Type	Ratio t_{i+1}/t_i
Linear	$t_i = i$	Arithmetic	$\rightarrow 1$ (constant additive step)
Quadratic	$t_i = i^2$	Polynomial, degree 2	$(i+1)^2/i^2 \rightarrow 1$ as $i \rightarrow \infty$
True Exponential	$t_i = b \cdot i^{i-1}$	Geometric (exponential)	b (constant multiplicative)
Fibonacci	$t_i = F_i$ (Fibonacci)	Sub-exponential	$\rightarrow \phi \approx 1.618$
Log-Spaced	$t_i = 10^{(i/8)}$	Geometric in log-space	$10^{(1/8)} \approx 1.334$ (constant)

B. Why True Exponential Fits Financial Volume

Empirical studies of financial market volume consistently find that the distribution of daily volume is approximately log-normal: $\log(V) \sim N(\mu, \sigma^2)$. This means the appropriate measure of 'how exceptional' a volume event is should be expressed on a logarithmic scale. A threshold system whose boundaries are equally spaced on a log axis — i.e., a geometric/exponential sequence — will therefore produce tiers that are equally probable under the log-normal model. Specifically, for a geometric threshold sequence with base b :

$$P(\text{Tier} = k) = \Phi\left(\frac{\log(t_{i+1}) - \mu}{\sigma}\right) - \Phi\left(\frac{\log(t_i) - \mu}{\sigma}\right)$$

When $t_i = b \cdot i^{i-1}$, the log-spacing $\log(t_{i+1}) - \log(t_i) = \log(b)$ is constant, and all tiers span equal intervals on the log-normal CDF. This means — in expectation — each tier is equally likely to occur, giving maximum information content per tier.

C. RVOL Lookback Selection

The RVOL lookback N controls the reference period. The optimal N minimises the variance of the RVOL estimator while tracking real changes in the volume regime. A heuristic: N should be long enough to span at least one full market cycle (accumulation $\rightarrow$ markup $\rightarrow$ distribution $\rightarrow$ markdown), which on daily charts is typically 20–100 bars. For intraday charts, N should span at least one full trading week (5–7 sessions $\times$ bars per session). The default $N=50$ is a practical compromise that works across most asset classes and timeframes.

D. Non-Repainting Guarantee

The pivot detection algorithm evaluates the candidate bar at offset L only after both the left window $[0, L-1]$ and right window $[L+1, 2L]$ have fully closed. At the current bar (offset 0), the earliest pivot that can be confirmed is at offset $2L$. No line is drawn until this confirmation is complete, ensuring the indicator does not repaint on bar close or during real-time bar formation.

$$\text{Earliest confirmed pivot} = \text{bar_index} - 2L$$

■ Lines extend to the right with `extend.right`, so they will appear to 'grow' as new bars form. This is not repainting — the anchor price and location are fixed at the moment of confirmation.